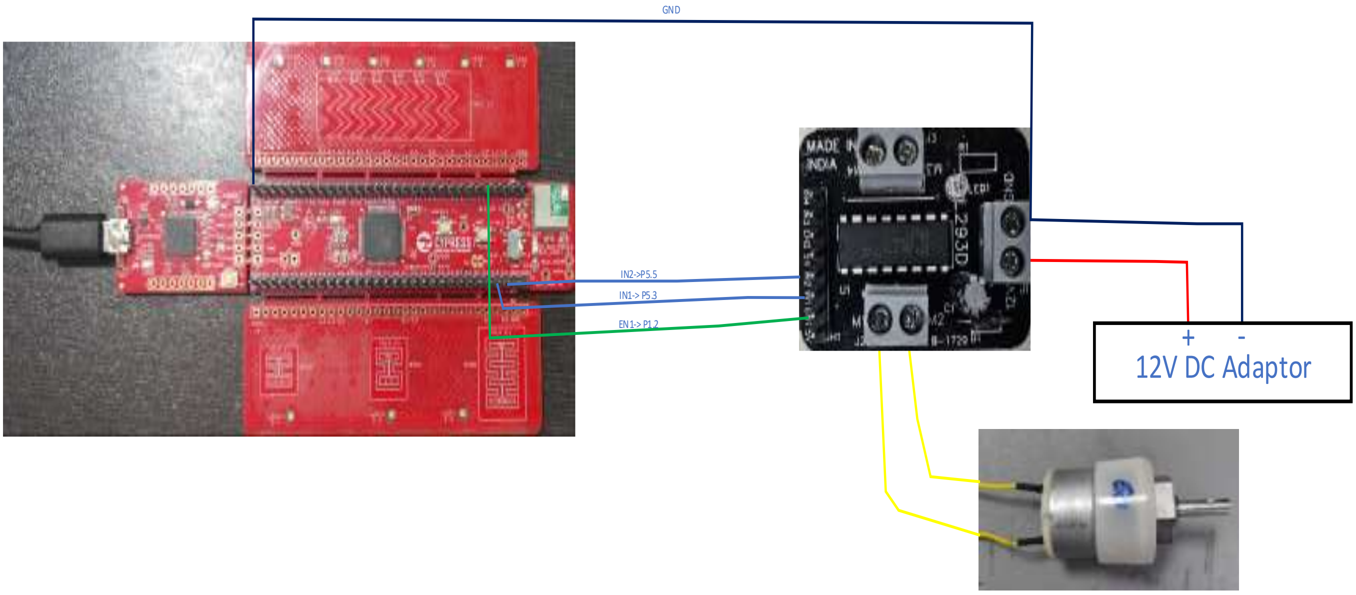
Motor connections with motor driver

Purpose: This write-up explains the test setup (hardware connections) required to interface the MCU with the Drive Board and the DC Motor.

Setup (Hardware connections):



<u>Description of various signals</u>

*IN1 and IN2 signals on the Drive board Control the direction of motor rotation.
Example: IN1=0, IN2=1 with PWM input on EN1 Motor turns left
IN1=1, IN2=0, with PWM input on EN1 Motor turns right
*EN1 Signal to control the speed of motor
Example: By varying the duty cycle of EN1(PWM) signal, the speed of the motor can be changed.
*12V and GND are DC supply connections.

<u>Sequence of Steps</u>

1. First, ensure that all power supplies are disconnected.
2. Next, interface the PSOC board with the Motor Drive Board as the following
   P5.3 --> IN1
   P5.5 --> IN2
   P1.2 --> EN1
3. Ensure all power supplies have a common return path. Do this by connecting the PSOC Ground with the Ground signal of the Motor Drive Board.
4. Connect the leads of the motor to M+ and M- signals of the Drive Board. Order does not matter.
5. Connect DC adaptor Positive and Negative to V+ and V- pins of the Motor drive Board. Ensure that the 12V power supply remains OFF while rigging up this connection.
6. Connect PSOC EVK to your PC using USB Cable.
7. Plug the 12DC adaptor to AC socket and power on.
8. Program the PSOC EVK with your PWM application and observe the motors rotate.

Exercises

a. Choose a direction of motor rotation. Change the duty cycle of the PWM gradually from 10% to 90% and watch the motor spin faster.

b. NEVER abruptly change the direction of rotation. The motor should first be brought to halt state by driving EN1=0 and waiting for a few milliseconds. With the motor coming to a full stop, the direction of rotation can now be changed. Done otherwise, the motor brushes may be permanently damaged.